

Total No. of Questions : 4]

SEAT No. :

PA-10062

[Total No. of Pages : 1

[6009] 353

T.E.(Information Technology) (Insem)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester-II) (314452)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume suitable data if necessary.*
- 4) *Attempt Q.1 or Q.2, Q3 or Q4.*

Q1) a) Explain 6V's for defining Big Data along with the factors responsible for data explosion? **[8]**

b) List and explain data processing infrastructure challenges in Big Data with suitable example. **[7]**

OR

Q2) a) List and explain choices for reengineering the data warehouse? **[8]**

b) Explain shared-everything and shared nothing architectures in detail with respect to Big data? **[7]**

Q3) a) Explain the following terms. **[7]**

- i) Expectation
- ii) Pair wise independence

b) Given that a person last purchase was coke. there is a 90% chance that his next purchase will also be coke. If a person's last purchase was Pepsi, there is an 80% chance that his next purchases will also Pepsi. **[8]**

- i) Given that a person is currently a Pepsi purchaser, what is the probability that he will purchase Coke two purchases from now?
- ii) Give that a person is currently a Coke drinker, what is the probability that he will purchase Pepsi three purchases from now?

OR

Q4) a) Explain Flajolet Martin Distance Sampling? Find the distinct element from the element stream 4,2,5,9,1,6,3,7. Consider the Hash function $h(x) = (3x+7) \bmod 32$. **[8]**

b) Find the variance and standard deviation for the following data set: 70, 60, 72, 42, 86 **[7]**



Total No. of Questions : 4]

SEAT No. :

PB145

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T.E. (I.T.) (Insem)

DATA SCIENCE & BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right indicate full marks.

Q1) a) “Data explosion plays an import role in Big Data” Justify the statement with proper explanation along with examples. **[8]**

b) Differentiate between Big Data and small Data with their processing architecture. **[7]**

OR

Q2) a) Define the relation between Artificial Intelligence, Statistical Learning and Machine Learning with sample example. **[8]**

b) Differentiate between Data ware house and Data mining with the Data Science perspective. **[7]**

Q3) a) In the Dark Ages, Harvard, Dartmouth and Yale admitted only male students. Assume that, at that time, 80 percent of the sons of Harvard men went to Harvard and the rest went to Yale, 40 percent of the sons of Yale men went to Yale; and the rest split evenly between Harvard and Dartmouth; and of the sons of Dartmouth men, 70 percent went to Dartmouth, 20 percent to Harvard, and 10 percent to Yale. **[8]**

i) Find the probability that the grandson of a man from Harvard went to Harvard.

ii) Modify the above by assuming that the son of a Harvard man always went to Harvard. Again, find the probability that the grandson of a man from Harvard went to Harvard.

b) Explain Blooms filter along with one application. Derive the equation for probability distribution in Blooms filter. **[7]**

OR

P.T.O.

Q4) a) Explain the Flajolet Mart in Distance Sampling. Find the distinct Element from the element stream $S=1,3,2,1,2,3,4,3,1,2,3,1$
 $h(x) = (6x+1) \bmod 5$. [8]

b) Only 1 in 1000 people have a rare disease A with $TP=0.99$ and $FP=0.02$. If one randomly tested individual is positive, what is the probability that they have a disease.

Given: $P(A)=0.001$, $P(A_0)=0.000$, $P(B/A)=0.99$, $P(B/A_0)=0.02$. [7]

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